

Addressing and Branch Instructions

Advanced Ladder Logic Logic and Limitations

1 Instruction Addressing

Every instruction in a PLC program must be assigned an address. This link is vital for the controller to understand the mapping between its software logic and the physical world.

Purpose of Addressing

Addressing serves to uniquely identify:

- **Input Mapping:** What physical PLC input is connected to which external input device (e.g., sensor, button).
- **Output Mapping:** What physical PLC output will drive which external load (e.g., motor starter, lamp).
- **Memory Linking:** It links the instruction to a specific **bit** in the data table portion of the processor's memory.

1.1 Standard Addressing Formats

Addressing depends on the PLC model and the number system used by the manufacturer:

- **Number Systems:** PLCs often utilize **Decimal**, **Octal**, or **Hexadecimal** addressing schemes.
- **Model Specifics:** For the **Allen-Bradley SLC 500**, addressing follows the **File:Slot/Terminal** format (e.g., I:1/0 or O:2/5).

2 Branch Instructions: Input Logic

Branching is used to create **OR logic** and complex combinations of input conditions.

- **Parallel Input Branching:** Allows multiple logic paths. At least one of the parallel branches must form a "TRUE" logical path for the rung to be energized.
- **OR Logic Execution:**
 - If *any* parallel path is continuous → Output = TRUE.
 - If *none* of the parallel paths are continuous → Output = FALSE.

3 Branch Instructions: Output Logic

Parallel output branching allows a single logic path to control multiple output devices simultaneously.

3.1 Standard Output Branching

Used when multiple loads must turn ON or OFF at the same time based on the same set of input conditions.

3.2 Conditional Output Branching

Additional input instructions can be programmed within the individual output branches. This allows for **enhanced conditional control**, where some outputs may be blocked or enabled depending on further logic, even while sharing the main rung conditions.

4 Nested Branches and Optimization

A nested branch is a structure that starts or ends within another branch.

- **Nested Architecture:** Used to create complex "if-else" or multi-layered logic paths.
- **Scan Efficiency:** Nested branches should be used carefully. In some processors, **eliminating nested contacts** (by repeating instructions to create parallel equivalents) can speed up the processor scan time and avoid redundant operations.

5 Programming Limitations

PLC manufacturers impose specific hardware and software constraints, often referred to as **Matrix Limitations**.

Common Constraints

Modern PLCs have finite limits on:

- **Series Elements:** The maximum number of instructions allowed in a single row.
- **Parallel Branches:** The maximum number of branches allowed within a single rung.
- **Output Rungs:** Limitations on how many outputs can be driven or branched together.

The "No Vertical Contact" Rule

A critical limitation in branch circuit programming is that **vertical contacts are not allowed**.

- **Flow Direction:** The processor scan logic always evaluates from left to right.
- **Restriction:** It never allows for current or logic flow to move back from **right-to-left**.

Systematic addressing and clean branching are the keys to professional PLC automation.